

1.Reverse a string without slicing:

"Hello World"

ANS

code:s="Hello World"

rev=""

for i in s:

rev= i+rev

print(rev)

output: dlrow olleH

x="Hello World"
x.reverse()

17
20

2.Count words in a sentence:

"A Quick Brown Fox Jump Over the Lazy Dog"

ANS

code:s = "A Quick Brown Fox Jump Over the Lazy Dog"

words = s.split()

print(len(words))

output: 9

1

3. Remove duplicate characters.

"Carrot"

ANS

code:s = "Carrot"

result = ""

for i in s:

if i not in result:

result += i

output:Carot

x="carrot"
x=set(x)

4. Difference between List and Set

ANS

List Set

Ordered collection Unordered collection

Duplicate values allowed Duplicate values not allowed

Uses [] brackets Uses {} brackets

Indexing possible Indexing not possible

1

5. Can sets contain lists?

ANS

No.

Because list is mutable (changeable) and set accepts only immutable values.

Reason:

Wrong:

s = {[1,2,3]}

Correct:

s = {(1,2,3)}

1

6. Why sets are faster?

ANS

- a. Sets use hashing ✓
- b. Searching is very fast ✓
- c. Duplicate values are automatically removed ✓

1

7. Difference between remove() and discard()

ANS

remove() ✓ discard() ✓

Gives error if item not found

Does not give error

Used when item surely exists

Safer to use ✓

Example:

s = {1,2,3}

s.remove(2)

s.discard(5)

1

8. How many methods to insert elements in the list? Total Method?

ANS

1. append() ✓

2. insert() ✓

3. extend() ✓

1

9. How many methods to delete elements in the lists?

ANS

1. remove() ✓

2. pop() ✓

3. clear() ✓

4. del ✓

pop(n)

1

10. How many Operators in Python Write their name in Explain?

ANS

IN python we have seven types of operator

1. Arithmetic Operators ✓

2. Comparison Operators ✓

3. Assignment Operators ✓

4. Logical Operators ✓

5. Bitwise Operators ✓

6. Membership Operators ✓

7. Identity Operators ✓

1

11. Write Precedence Rule of arithmetic in Python ?

ANS

1. Parentheses () ✓

2. Exponent ** ✓

3. Multiplication/Division * / // % ✓

4. Addition/Subtraction + - ✓

1

12. How to sort elements in list?

[5,11,14,7,8,3,2,5,9]

ANS

l = [5,11,14,7,8,3,2,5,9]

l.sort()

print(l)

Output:

[2,3,5,5,7,8,9,11,14]



13. What is Dictionary?

ANS

Dictionary stores data in key-value pairs.

Example:

d = {"name": "Damini", "age": 21}



14. How many methods to create dictionary?

ANS

Total method to create dictionary are three:

1. Using {}

2. Using dict()

3. Using from keys()



15. How many methods to access value in dictionary?

ANS

Total method to accessing values in dictionary are five :

1. Using key

2. get()

3. keys()

4. values()

5. items()

index



16. How many methods to delete elements in dictionary?

ANS

Total methods to delete elements in dictionary are four:

1. pop()

2. pop item()

3. clear()

4. del



17. How to perform mathematical operations in sets write method name?

ANS

1. union()

2. intersection()

3. difference()

4. symmetric difference()



Example:

a = {1,2,3}

b = {3,4,5}

print(a. Union(b))

print(a. Intersection(b))

18. What is packing variable in tuple? Explain using example?

ANS

Packing means storing multiple values in one tuple.

Example:

t = 1,2,3

print(t)

Output:

(1,2,3)

19. What is unpacking variables in tuple? Explain using example?

ANS

Unpacking means storing tuple values into separate variables.

Example:

t = (1,2,3)

a,b,c = t

print(a)

print(b)

print(c)

20. Difference between Set and Dictionary

ANS

Set

Stores only values

No keys

Duplicate values not allowed

Dictionary

Stores key-value pairs

Keys are compulsory

Duplicate keys not allowed

key-value pairs